Chapter 2

Limits and Their Duals

2.1 Terminal and Initial Objects

Definition 2.1.1 (Terminal Object)**.** A *terminal object* 1 of a category $\mathcal{C}$ is an object such that for any object A, there is a unique morphism $!_A : A \longrightarrow 1$.

A category may have any number of terminal objects.

Theorem 2.1.2. A terminal object is unique up to isomorphism.

Proof. Suppose that 1 and $1'$ are terminal objects in a category $\mathcal{C}$. On one hand, since 1 is a terminal object, there is a unique morphism $!_{1'} : 1' \longrightarrow 1$. On the other hand, since $1'$ is a terminal object, there is a unique morphism $!_1 : 1 \longrightarrow 1'$.

$$1' \underset{!_1}{\overset{!_{1'}}{\rightleftarrows}} 1$$

Then the composite $!_{1'} \circ !_1 : 1 \longrightarrow 1' \longrightarrow 1$ equals id_1, since the morphism to 1 is unique. In the same way, the composite $!_1 \circ !_{1'} : 1' \longrightarrow 1 \longrightarrow 1'$ equals $id_{1'}$, since the morphism to $1'$ is also unique. Therefore, $!_1$ and $!_{1'}$ are mutually inverse and $1 \cong 1'$. □

Definition 2.1.3 (Global Element)**.** A *global element* of an object A in a category $\mathcal{C}$ is any morphism $1 \longrightarrow A$.

Definition 2.1.4 (Initial Object)**.** An *initial object* $\emptyset$ of a category $\mathcal{C}$ is an object such that for any object C, there is a unique morphism $!_C : \emptyset \longrightarrow C$.

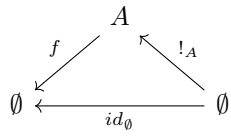
A category may have any number of initial objects.

Theorem 2.1.5. An initial object is unique up to isomorphism.

Proof. By the dual of the proof of Theorem 2.1.2. □

Theorem 2.1.6. For any object A and morphism $f : A \longrightarrow \emptyset$, f is split epic.

Proof. Consider a morphism $f \circ !_A : \emptyset \longrightarrow A \longrightarrow \emptyset$. Since there is only one morphism from $\emptyset$ to any object, one has $f \circ !_A = id_\emptyset$. Thus f has a right inverse $!_A : \emptyset \longrightarrow A$ and is split epic.



□

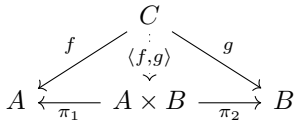
Exercise 2.1.7. Suppose that there is a terminal object 1. Show that every morphism $f : 1 \longrightarrow B$ is monic.

Exercise 2.1.8. An object which is both terminal and initial is called a *zero object*. Show that, if there is an initial object $\emptyset$, a terminal object 1, and a morphism $1 \longrightarrow \emptyset$, then 1 and $\emptyset$ are zero objects.

2.2 Products and Coproducts

Definition 2.2.1 (Binary Product)**.** A *binary product* of objects A and B in $\mathcal{C}$ is specified by:

- an object $A \times B$ of $\mathcal{C}$
- two *projections*: morphisms $\pi_1 : A \times B \longrightarrow A$ and $\pi_2 : A \times B \longrightarrow B$, such that given any object C and morphisms $f : C \longrightarrow A$, $g : C \longrightarrow B$, there is a unique morphism $\langle f, g \rangle : C \longrightarrow A \times B$ for which $\pi_1 \circ \langle f, g \rangle = f$ and $\pi_2 \circ \langle f, g \rangle = g$, i.e. the following diagram commutes.

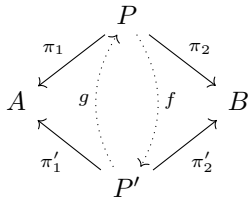


We shall simply refer to a binary product $A \times B$ instead of the triple $\langle A \times B, \pi_1, \pi_2 \rangle$.

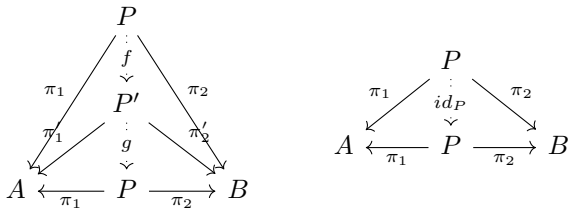
Exercise 2.2.2. Show that $\langle f, g \rangle$ is monic if f is monic.

Theorem 2.2.3. For any objects A and B, a binary product of A and B is unique up to isomorphism if it exists.

Proof. Suppose that P and P' are binary products of A and B. By Definition 2.2.1, there is a unique morphism $f : P \longrightarrow P'$ such that $\pi'_1 \circ f = \pi_1$ and $\pi'_2 \circ f = \pi_2$, and there is a unique morphism $g : P' \longrightarrow P$ such that $\pi_1 \circ g = \pi'_1$ and $\pi_2 \circ g = \pi'_2$.



Therefore, the composite $g \circ f$ is a morphism from P to P that makes the left diagram below commute, namely, $\pi_1 \circ (g \circ f) = (\pi_1 \circ g) \circ f = \pi_1' \circ f = \pi_1$ and $\pi_2 \circ (g \circ f) = (\pi_2 \circ g) \circ f = \pi_2' \circ f = \pi_2$.



On the other hand, the right diagram above commutes as well. Since P is a binary product of A and B, $g \circ f = id_P$. Similarly, $f \circ g = id_{P'}$. Thus, g and f are mutually inverse, and $P \cong P'$. □

Theorem 2.2.4. For any objects A, B, $A \times B \cong B \times A$.

Proof. Both $\langle A \times B, \pi_1, \pi_2 \rangle$ and $\langle B \times A, \pi_2, \pi_1 \rangle$ are binary products of A and B. Therefore, by Theorem 2.2.3, one has $A \times B \cong B \times A$. □

Exercise 2.2.5. Show that the product constructions are associative: for any objects A, B, C, $A \times (B \times C) \cong (A \times B) \times C$.

Definition 2.2.6 (Product of Morphisms)**.** Let $\mathcal{C}$ be a category with binary products and take morphisms $f : A \longrightarrow B$ and $f' : A' \longrightarrow B'$. We write $f \times f' : A \times A' \longrightarrow B \times B'$ for the *product of morphisms* $\langle f \circ \pi_1, f' \circ \pi_2 \rangle$, where $\pi_1 : A \times A' \longrightarrow A$ and $\pi_2 : A \times A' \longrightarrow A'$ are the projections. In other words, $f \times f'$ is the unique morphism that makes the following diagram commute.

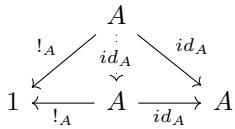
$$\begin{array}{ccccc} A & \xleftarrow{\pi_1} & A \times A' & \xrightarrow{\pi_2} & A' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f \times f'} & & \downarrow{\scriptstyle f'} \\ B & \xleftarrow[\pi_1]{} & B \times B' & \xrightarrow[\pi_2]{} & B' \end{array}$$

Exercise 2.2.7. Show that each of the following equations hold ($f : A \longrightarrow B,\ f' : A \longrightarrow B',\ g : B \longrightarrow C$, and $g' : B' \longrightarrow C'$).

$$id_A \times id_{A'} = id_{A \times A'}$$
$$(g \times g') \circ (f \times f') = g \circ f \times g' \circ f'$$
$$(g \times g') \circ \langle f, f' \rangle = \langle g \circ f, g' \circ f' \rangle$$

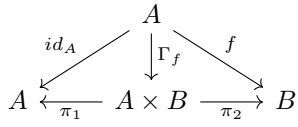
Theorem 2.2.8. In a category $\mathcal{C}$ with binary products, any object A is isomorphic to $1 \times A$ ($A \cong 1 \times A$).

Proof. Consider any morphism $f : A \longrightarrow A$ such that $id_A = id_A \circ f$, which means $f = id_A$. Thus, id_A is the unique morphism that makes the following diagram commute.

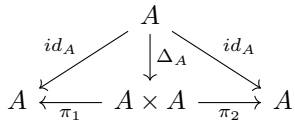


Therefore, $\langle A, !_A, id_A \rangle$ is a binary product of 1 and A. By Theorem 2.2.3, $A \cong 1 \times A$. □

Definition 2.2.9 (Graph)**.** Suppose that objects A and B have a product $A \times B$. Take any morphism $f : A \longrightarrow B$, the *graph of* f is defined as the morphism Γ_f that makes the following diagram commute:



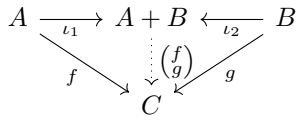
Definition 2.2.10 (Diagonal)**.** For any object A, the *diagonal* $\Delta_A : A \longrightarrow A \times A$ is a graph of id_A.



Exercise 2.2.11. Show that, if a category $\mathcal{C}$ has binary products, $\langle h, k \rangle = (h \times k) \circ \Delta_T$ for any $h : T \longrightarrow A$ and $k : T \longrightarrow B$.

Definition 2.2.12 (Binary Coproduct)**.** A *binary coproduct* of objects A and B in $\mathcal{C}$ is specified by:

- an object $A + B$ of $\mathcal{C}$
- two projections: morphisms $\iota_1 : A \longrightarrow A + B$ and $\iota_2 : B \longrightarrow A + B$, such that given any object C and morphisms $f : A \longrightarrow C$, $g : B \longrightarrow C$, there is a unique morphism $\binom{f}{g} : A + B \longrightarrow C$ for which $\binom{f}{g} \circ \iota_1 = f$ and $\binom{f}{g} \circ \iota_2 = g$, i.e. the following diagram commutes.



Theorem 2.2.13. For any objects A and B, a binary coproduct of A and B is unique up to isomorphism if it exists.

Theorem 2.2.14. In a category $\mathcal{C}$ with binary coproducts, any object A is isomorphic to $\emptyset + A$ ($A \cong \emptyset + A$).

Theorem 2.2.15. For any objects A, B, $A + B \cong B + A$.

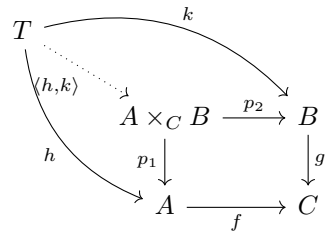
Proof. Reverse the arrows in the proofs of Theorem 2.2.3, Theorem 2.2.8, and Theorem 2.2.4, respectively. □

Exercise 2.2.16. Show the dual of Exercise 2.2.5, namely, that the coproduct constructions are associative: for any objects A, B, C, $A+(B+C) \cong (A+B)+C$.

2.3 Pullbacks and Pushouts

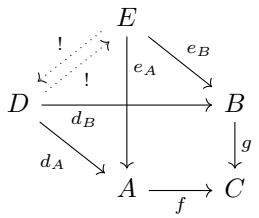
Definition 2.3.1 (Pullback)**.** A *pullback* for a pair of morphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$ in a category $\mathcal{C}$ is given by:

- an object $A \times_C B$ of $\mathcal{C}$
- a pair of morphisms $p_1 : A \times_C B \longrightarrow A$ and $p_2 : A \times_C B \longrightarrow B$ (called *projection arrows*) for which $f \circ p_1 = g \circ p_2$ such that given any object T and any pair of morphisms $h : T \longrightarrow A$ and $k : T \longrightarrow B$ in $\mathcal{C}$ for which $f \circ h = g \circ k$, there is a unique morphism from T to $A \times_C B$, denoted by $\langle h, k \rangle$ (see Remark 2.4.6), for which $p_1 \circ \langle h, k \rangle = h$ and $p_2 \circ \langle h, k \rangle = k$, i.e. the following diagram commutes.

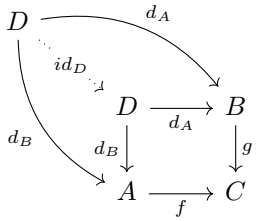


Theorem 2.3.2. A pullback for a given pair of morphisms is determined up to isomorphism.

Proof. Suppose that $D, d_A : D \longrightarrow A, d_B : D \longrightarrow B$ and $E, e_A : E \longrightarrow A, e_B : E \longrightarrow B$ are pullbacks for a pair of morphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$. By the definition of pullback, there is a unique morphism from D to E and vice versa.

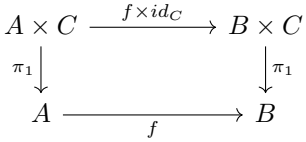


On the other hand, since D is a pullback, the morphism from D to D that makes the following diagram commute is unique, that is, id_D.

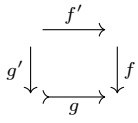


This means that the morphism $D \xrightarrow{!} E \xrightarrow{!} D$ is id_D. Similarly, the morphism $E \xrightarrow{!} D \xrightarrow{!} E$ is id_E. Hence $D \cong E$. □

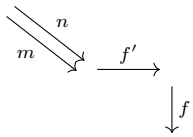
Exercise 2.3.3. Show that, for any morphism $f : A \longrightarrow B$ and object C, if A and B have a product with C, then the following square is a pullback.



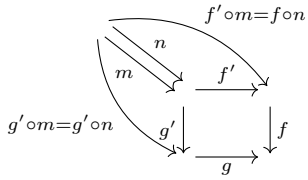
Theorem 2.3.4. If the following diagram is a pullback and g is monic, then f' is also monic.



Proof. For any two morphisms m, n such that $f' \circ m = f' \circ n$, we have $f \circ f' \circ m = f \circ f' \circ n$.

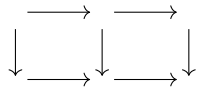


On the other hand, $f \circ f' = g \circ g'$. So we also have $g \circ g' \circ m = g \circ g' \circ n$. This means that the following diagram commutes.

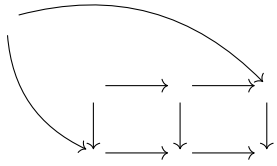


Since the square is a pullback, $m = n$. □

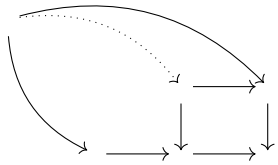
Theorem 2.3.5. In the following diagram, the outer rectangle is a pullback if the two inner squares are pullbacks.



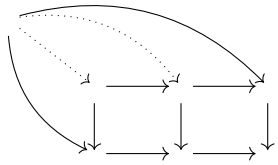
Proof. Consider any two morphisms that make the following diagram commute.



Since the right square is a pullback, there is a unique morphism to the upper left corner of it.

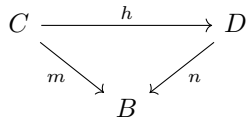


Then, since the left square is also a pullback, there is a unique morphism to the upper left corner of it. Therefore, the whole square is a pullback.

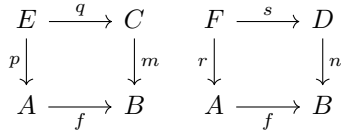


□

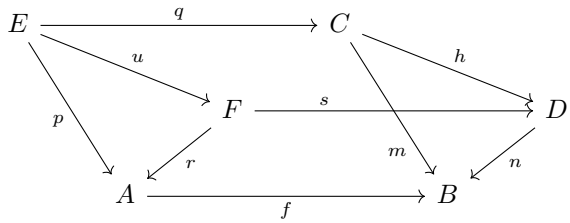
Exercise 2.3.6. Suppose that there is a commutative triangle



and suppose that m and n have the following pullbacks along f:



Show that, then, there is a unique $u : E \longrightarrow F$ that makes the following diagram commute, and E, u, q a pullback for h and s:

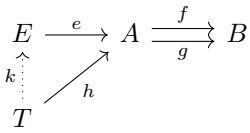


2.4 Equalizers and Coequalizers

Definition 2.4.1 (Equalizer)**.** A morphism $e : E \longrightarrow A$ is an *equalizer* for a pair of morphisms $f, g : A \longrightarrow B$ in a category $\mathcal{C}$ iff

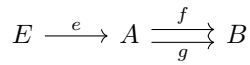
- $f \circ e = g \circ e$
- given any $h : T \longrightarrow A$ with $f \circ h = g \circ h$, there is a unique $k : T \longrightarrow E$ for which $h = e \circ k$.

i.e. the following diagram commutes.

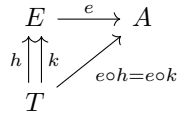


Theorem 2.4.2. Every equalizer is monic.

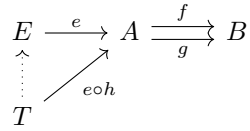
Proof. Suppose that e is an equalizer of f and g in the following diagram, i.e. $f \circ e = g \circ e$.



Consider two morphisms h and k such that $e \circ h = e \circ k$.



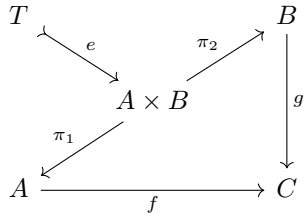
Since $f \circ e = g \circ e$, it follows that $f \circ e \circ h = g \circ e \circ h$. Then there is a unique morphism from T to E that makes the following diagram commute.



Since $e \circ h = e \circ k$, it turns out that $h = k$. Therefore e is monic. □

Theorem 2.4.3. If a category has all binary products and equalizers for any two morphisms, then it has a pullback for any two morphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$.

Exercise 2.4.4. Prove Theorem 2.4.3 by showing that the equalizer for $f \circ \pi_1$ and $g \circ \pi_2$ gives a pullback diagram for $f : A \longrightarrow C$ and $g : B \longrightarrow C$.



Exercise 2.4.5. Suppose that a category has a pullback for every pair $f : A \longrightarrow C$ and $g : B \longrightarrow C$ and all binary products. Then it has a morphism from $A \times_C B$ to $A \times B$ and it is an equalizer for $f \circ \pi_1$ and $g \circ \pi_2$.

Remark 2.4.6. Exercise 2.4.5 justifies the notation $\langle h, k \rangle$ in Definition 2.3.1, which is borrowed from that for binary products. Suppose that $\mathcal{C}$ has binary products. Then $A \times_C B$ is an equalizer of $f \circ \pi_1$ and $g \circ \pi_2$, so there is a monic $e : A \times_C B \rightarrowtail A \times B$ (Theorem 2.4.2) with $p_1 = \pi_1 \circ e$ and $p_2 = \pi_2 \circ e$. For any $h : T \longrightarrow A$ and $k : T \longrightarrow B$ with $f \circ h = g \circ k$, the pairing $\langle h, k \rangle : T \longrightarrow A \times B$ satisfies $f \circ \pi_1 \circ \langle h, k \rangle = g \circ \pi_2 \circ \langle h, k \rangle$, hence factors uniquely through e, and the morphism $T \longrightarrow A \times_C B$ so obtained is exactly the unique morphism in Definition 2.3.1. The two readings of $\langle h, k \rangle$ thus differ only by composition with e. In $\mathcal{S}et$, for instance, $A \times_C B = \{(a, b) \in A \times B \mid f(a) = g(b)\}$ and $\langle h, k \rangle(t) = (h(t), k(t))$ in either reading.

Theorem 2.4.7. The graph of $f : A \longrightarrow B$ is an equalizer for $f \circ \pi_1 : A \times B \longrightarrow B$ and $\pi_2 : A \times B \longrightarrow B$.

Exercise 2.4.8. Prove Theorem 2.4.7.

2.5 Kernel and Cokernel Pairs

Definition 2.5.1 (Kernel Pair)**.** For any morphism $f : A \longrightarrow B$, a *kernel pair* for f is the pair of projection arrows in a pullback of f along itself.

$$\begin{array}{ccc} K & \xrightarrow{h} & A \\ {\scriptstyle k}\downarrow & & \downarrow{\scriptstyle f} \\ A & \xrightarrow[f]{} & B \end{array}$$

The dual notion of a kernel pair is a *cokernel pair*.

Definition 2.5.2 (Cokernel Pair)**.** For any morphism $f : A \longrightarrow B$, a *cokernel pair* for f is the pair of projection arrows in a pushout of f along itself.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle h} \\ B & \xrightarrow[k]{} & C \end{array}$$

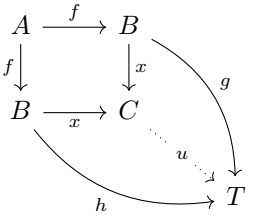
Theorem 2.5.3. Kernel pairs for any morphism are determined up to isomorphism.

Proof. It follows directly from Theorem 2.3.2. □

Theorem 2.5.4. If a cokernel pair for a morphism f consists of a single morphism taken twice, then f is epic.

Proof. Suppose that the pair $x, x : B \longrightarrow C$ is a cokernel pair for $f : A \longrightarrow B$ and two morphisms $g, h : B \longrightarrow T$ are such that $g \circ f = h \circ f$. Then there must be a unique morphism from C to T (say u) and $g = u \circ x = h$. Therefore f is

epic.



□

2.6 Cones and Cocones

Definition 2.6.1 (Cone)**.** (McLarty, 1992, p.48) Consider a diagram D with an object A_i at each vertex i, and a morphism f_e at each edge e. A *cone over* D consists of an object C and morphisms $p_i : C \longrightarrow A_i$ to each vertex such that, for every edge e with morphism $f_e : A_i \longrightarrow A_j$ we have $f_e \circ p_i = p_j$.

Definition 2.6.2 (Limit)**.** A *limit* for the diagram D is a cone over D (an object C and morphisms p_i) which satisfy the following properties:

- For any cone (say, T and q_i) over D, there is a unique morphism $u : T \longrightarrow C$, such that $p_i \circ u = q_i$ for each vertex i.

The morphisms p_i are called *projection arrows*.

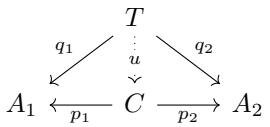
Example 2.6.3. Consider a diagram which has no vertices. Then a cone over it is a limit iff it is a terminal object.

$$T \overset{u}{\dashrightarrow} C$$

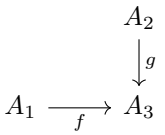
Example 2.6.4. Consider a diagram which has two vertices and no edges.

$$A_1 \qquad\qquad A_2$$

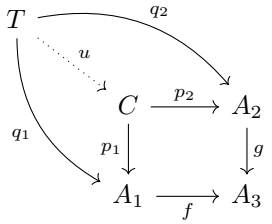
Then a cone over it is a limit iff it is a product diagram.



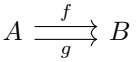
Example 2.6.5. Consider a diagram which has three vertices and two edges.



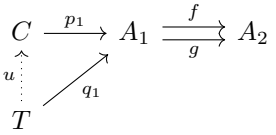
Then a cone over it is a limit iff it is a pullback diagram.



Example 2.6.6. Consider a diagram which has two vertices and two edges between them.



Then a cone over it is a limit iff it is an equalizer diagram.



Cocone is the dual notion of a cone, and *colimits* are the dual notion of limits.

Definition 2.6.7 (Cocone)**.** (McLarty, 1992, p.50) A *cocone* on the diagram D is an object C together with morphisms $q_i : A_i \longrightarrow C$ for each vertex i, such that for every edge e with morphism $f_e : A_i \longrightarrow A_j$ we have $q_j \circ f_e = q_i$.

Definition 2.6.8 (Colimit)**.** (McLarty, 1992, p.50) A *colimit* for the diagram D is a cocone on D, namely C with morphisms p_i, with the following property: given any cocone T, $r_i : A_i \longrightarrow T$ on D, there is exactly one morphism $u : C \longrightarrow T$ such that for every vertex $i, u \circ p_i = r_i$.

2.7 All Finite Limits and Colimits

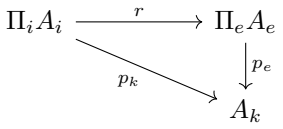
Definition 2.7.1 (Finite Products)**.** A category $\mathcal{C}$ has *finite* products iff it has products for every finite diagram.

Definition 2.7.2 (All Finite Limits)**.** A category $\mathcal{C}$ has *all finite limits* iff $\mathcal{C}$ has a limit for every finite diagram.

Theorem 2.7.3. If a category has a terminal object, binary products, and an equalizer for every parallel pair of arrows, then it has all finite limits.

Proof. Suppose that there is a product $\Pi_i A_i$ of all the A_i, with projection $p_j : \Pi_i A_i \longrightarrow A_j$ for each vertex j. Use the notation A_e for the codomain of the morphism at edge e, so that if $f_e : A_j \longrightarrow A_k$ then $A_e = A_k$. Suppose that there is a product $\Pi_e A_e$ with one factor A_e for each edge e, and a projection $p_e : \Pi_e A_e \longrightarrow A_k$ for each $f_e : A_j \longrightarrow A_k$.

Let $r : \Pi_i A_i \longrightarrow \Pi_e A_e$ be the morphism such that, for each edge e with $f_e : A_j \longrightarrow A_k$, the following triangle commutes:



Let $s : \Pi_i A_i \longrightarrow \Pi_e A_e$ be the morphism such that for each edge e the following square commutes:

$$\begin{array}{ccc} \Pi_i A_i & \xrightarrow{s} & \Pi_e A_e \\ {\scriptstyle p_j}\downarrow & & \downarrow{\scriptstyle p_e} \\ A_j & \xrightarrow[f_e]{} & A_k \end{array}$$

Then the diagram D has a limit iff there is an equalizer for r and s. More specifically, $l : L \longrightarrow \Pi_i A_i$ is an equalizer for r and s iff L together with all $p_i \circ l : L \longrightarrow A_i$ forms a limit cone for D.

Consider any cone $T, h_i : T \longrightarrow A_i$ over D. There is a unique $h : T \longrightarrow \Pi_i A_i$ such that for each vertex k we have $p_k \circ h = h_k$. Then $r \circ h = s \circ h$ iff for each edge e, $p_e \circ r \circ h = p_e \circ s \circ h$. But, by the definition of r, s and h, that last equation means that $h_k = f_e \circ h_j$, where $f_e : A_j \longrightarrow A_k$. Thus T, h factors uniquely through L, l iff the cone T, h_i factors uniquely through the cone $L, p_i \circ l$. □

Theorem 2.7.4. If a category has a terminal object and pullbacks for all corner of arrows, then it has all finite limits.

Proof. By Theorem 2.7.3, Theorem 2.7.5 and Theorem 2.7.7. □

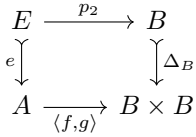
Theorem 2.7.5. A category with a terminal object and pullbacks for all corners of arrows, has a product for each pair of objects.

Exercise 2.7.6. Prove Theorem 2.7.5 by showing that $\langle A \times_1 B, p_1, p_2 \rangle$ in the following diagram is also a product diagram for A and B.

$$\begin{array}{ccc} A \times_1 B & \xrightarrow{p_2} & B \\ {\scriptstyle p_1}\downarrow & & \downarrow{\scriptstyle !_B} \\ A & \xrightarrow[!_A]{} & 1 \end{array}$$

Theorem 2.7.7. A category with a product for each pair of objects, and a pullback for every corner of arrows, has an equalizer for every parallel pair.

Exercise 2.7.8. Prove Theorem 2.7.7 by showing that the following e is an equalizer for any pair $f, g : A \longrightarrow B$.


E
p2
B
e
ΔB
A
⟨f,g⟩
B × B

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